

Segmenter: Transformer for Semantic Segmentation

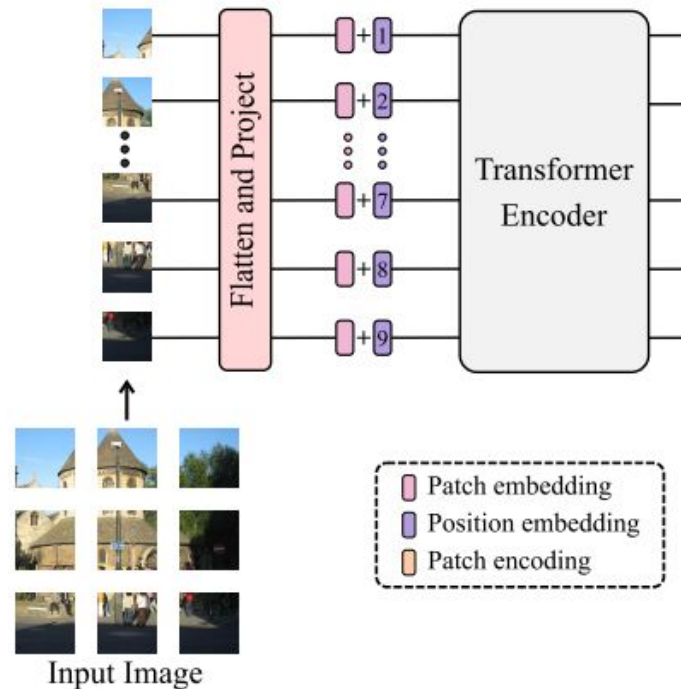
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Introduction

- SOTA methods deploy FCN
- local nature of convolutions limits the access to the global information in the image
- several segmentation methods use attention mechanism but still rely on convolutions
- semantic segmentation as a **sequence-to-sequence** problem using a transformer architecture

Patch embedding and positional encoding

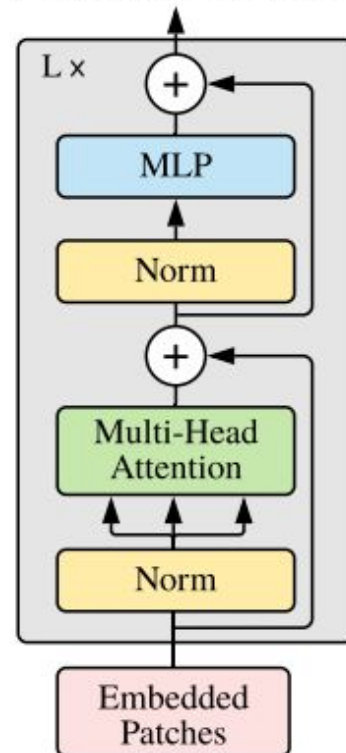
1. the image is splitted into a sequence of N patches of size $P \times P \times C$
2. the $3D$ patches are flattened into $1D$ vectors
3. each flattened patch is linearly projected using a learnable weight matrix
4. to the resulting projected matrix a learnable positional encoding matrix is added



Transformer encoder stack

- stack of L transformer encoder layers
- each layer is composed of 2 sub-layers:
 - $a_{i-1} = \text{MHSA}(\text{LayerNorm}(z_{i-1}) + z_{i-1})$
 - $z_i = \text{MLP}(\text{LayerNorm}(a_{i-1}) + a_{i-1})$
- outputs a contextualized encoding sequence $\mathbf{z}_L$ containing rich semantic information

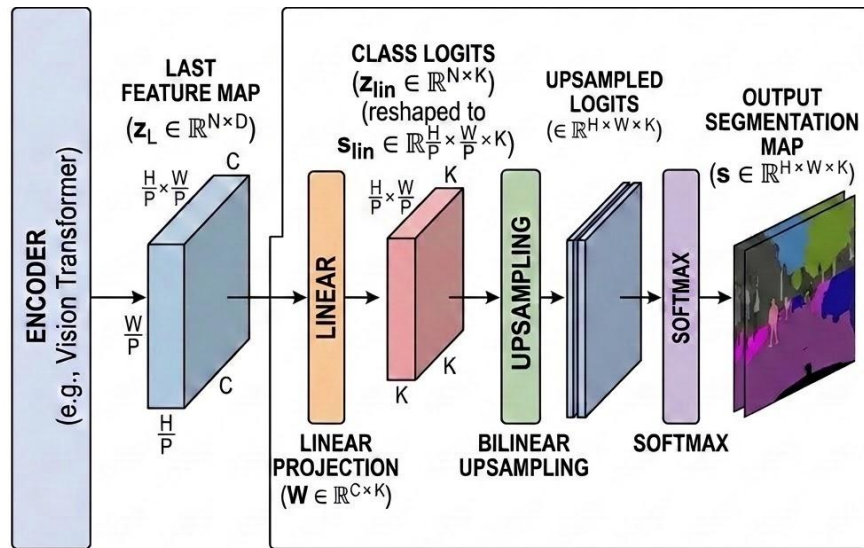
Transformer Encoder



Decoder variant 1: Linear

point-wise linear layer applied to patch encodings z_L :

1. produce patch-level class logits z_{lin}
2. reshape into a 2D feature map s_{lin}
3. bilinearly upsample to the original image size s
4. softmax on the class dimension to obtain the final segmentation map

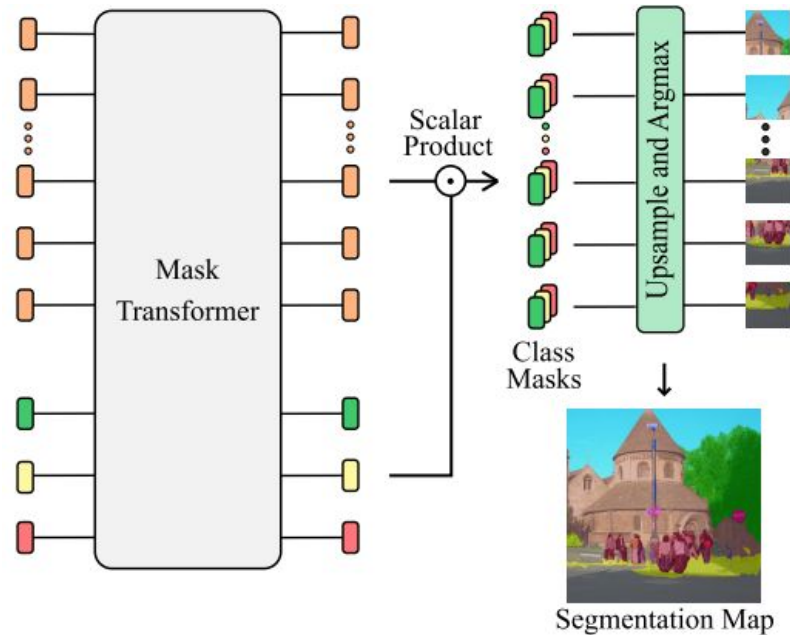


N : number of patches ($\frac{H}{P} \times \frac{W}{P}$)

P : patch size
 H, W : image height and width
 D : encoding channel dimension
 K : number of semantic classes

Decoder variant 2: Mask transformer

1. K learnable class embeddings $cls = [cls_1, \dots, cls_K]$ size D
2. M transformer encoder layers
3. compute $Masks(z'_M, C) = z'_M C^T$
4. reshape into a 2D matrix and bilinearly upsample into a $H \times W \times K$ feature map
5. compute softmax to output the final semantically segmented image



Implementation details

- **different models** of different sizes
- **per-pixel cross-entropy loss**
- **data augmentation**: mean subtraction, random resizing and random left-right flipping
- **optimization**: SGD and poly learning rate decay
- **inference**: sliding-window with a resolution matching the training size

Model	Backbone	Layers	Token size	Heads	Params
Seg-Ti	ViT-Ti	12	192	3	6M
Seg-S	ViT-S	12	384	6	22M
Seg-B	ViT-B	12	768	12	86M
Seg-B [†]	DeiT-B	12	768	12	86M
Seg-L	ViT-L	24	1024	16	307M

Ablation study: regularization and dataset size

		Stochastic Depth		
		0.0	0.1	0.2
Dropout	0.0	45.01	45.37	45.10
	0.1	42.02	42.30	41.14
	0.2	36.49	36.63	35.67

Performance comparison of Seg-S/16 models trained with increasing dataset size and evaluated on ADE20K validation set.

Mean IoU comparison of different regularization schemes using Seg-S/16 on ADE20K validation set

Dataset Size	4k	8k	12k	16k	20k
mIoU (SS)	38.31	41.87	43.42	44.61	45.37

Ablation study: model size and patch size

Method	Backbone	Patch size	Im/sec	ImNet acc.	mIoU (SS)
Seg-Ti/16	ViT-Ti	16	396	78.6	39.03
Seg-S/32	ViT-S	32	1032	80.5	40.64
Seg-S/16	ViT-S	16	196	83.7	45.37
Seg-B [†] /16	DeiT-B	16	92	85.2	47.08
Seg-B/32	ViT-B	32	516	83.3	43.07
Seg-B/16	ViT-B	16	92	86.0	48.06
Seg-B/8	ViT-B	8	7	85.7	49.54
Seg-L/16	ViT-L	16	33	87.1	50.71



(a) Patch size 32×32

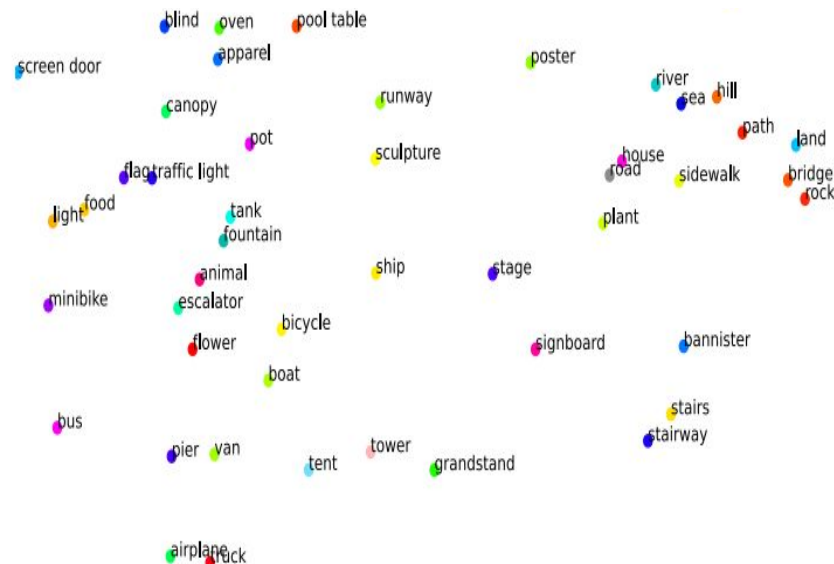
(b) Patch size 16×16

(c) Patch size 8×8

(d) Ground Truth

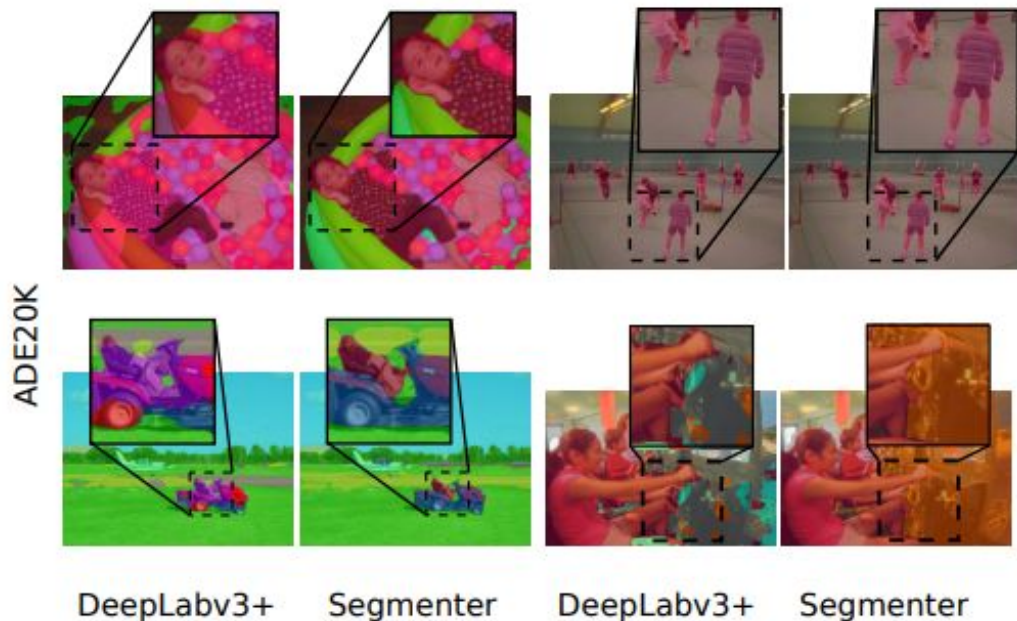
Ablation study: decoder variant

Method	Decoder	Small	Medium	Large	mIoU (SS)
DeepLab RNeSt-101	UNet	37.85	50.89	50.67	46.47
Seg-B/32	Linear	31.95	47.82	49.44	43.07
Seg-B-Mask/32	Mask	32.29	49.44	50.82	44.19
Seg-B [†] /16	Linear	38.31	50.91	52.08	47.10
Seg-B [†] -Mask/16	Mask	40.49	51.37	54.24	48.70
Seg-B/16	Linear	39.57	51.32	53.28	48.06
Seg-B-Mask/16	Mask	40.16	52.61	52.66	48.48
Seg-B/8	Linear	41.43	54.35	52.85	49.54
Seg-L/16	Linear	42.08	54.67	55.39	50.71
Seg-L-Mask/16	Mask	42.02	54.83	57.06	51.30



Ablation study: SOTA comparison on ADE20K

Method	Backbone	Im/sec	mIoU	+MS
OCR [60]	HRNetV2-W48	83	-	45.66
ACNet [24]	ResNet-101	-	-	45.90
DNL [57]	ResNet-101	-	-	45.97
DRANet [22]	ResNet-101	-	-	46.18
CPNet [58]	ResNet-101	-	-	46.27
DeepLabv3+ [10]	ResNet-101	76	45.47	46.35
DeepLabv3+ [10]	ResNeSt-101	15	46.47	47.27
DeepLabv3+ [10]	ResNeSt-200	-	-	48.36
SETR-L MLA [67]	ViT-L/16	34	48.64	50.28
Swin-L UperNet [35]	Swin-L/16	34	52.10	53.50
Seg-B [†] /16	DeiT-B/16	77	47.08	48.05
Seg-B [†] -Mask/16	DeiT-B/16	76	48.70	50.08
Seg-L/16	ViT-L/16	33	50.71	52.25
Seg-L-Mask/16	ViT-L/16	31	51.82	53.63

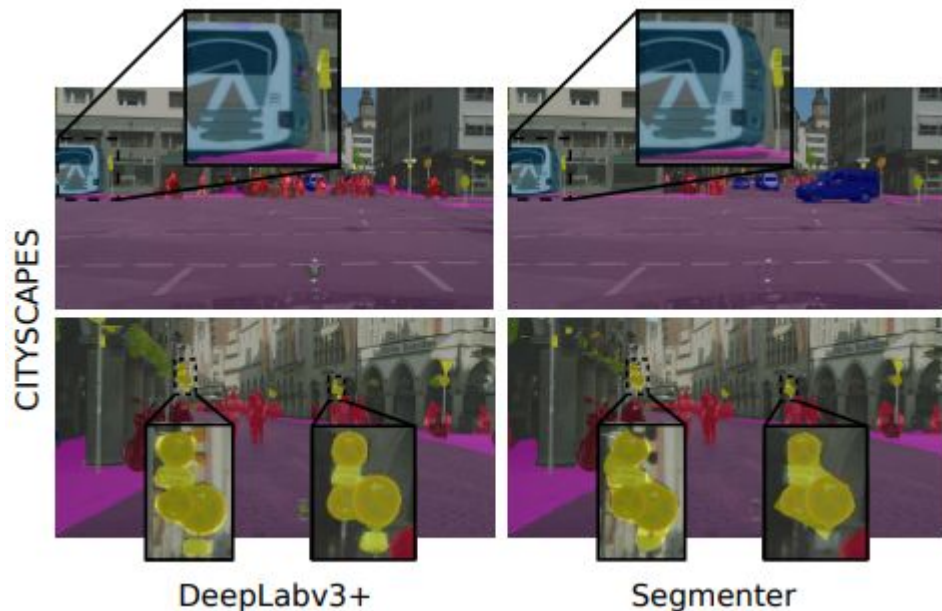


Ablation study: SOTA comparison on Pascal context

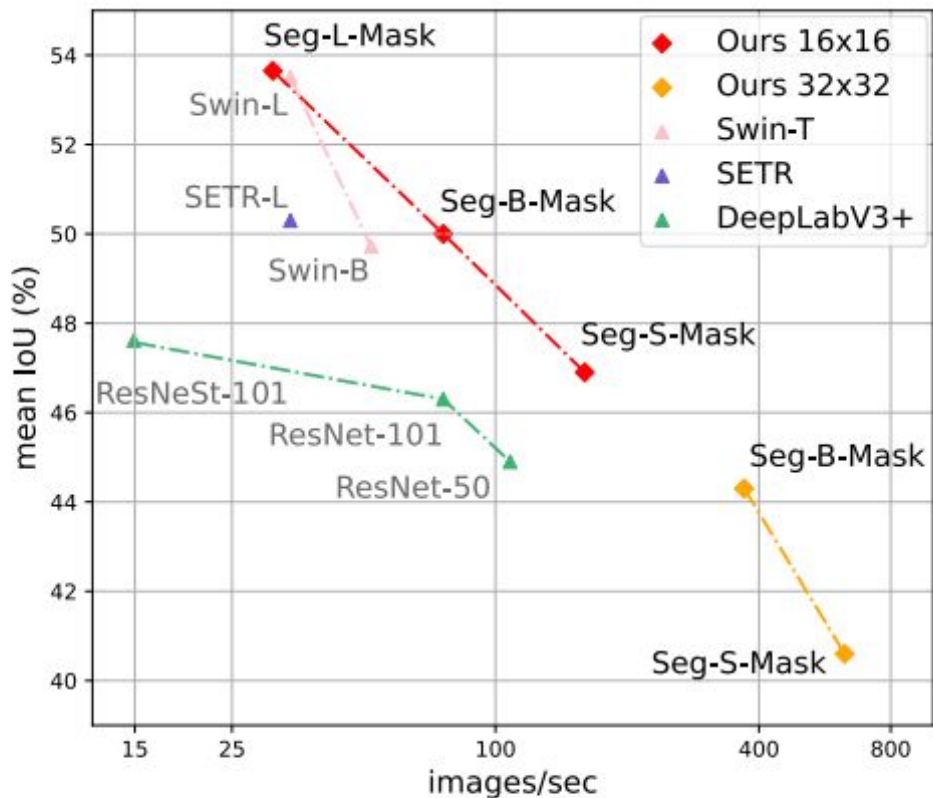
Method	Backbone	mIoU (MS)
DeepLabv3+ [10]	ResNet-101	48.5
DANet [23]	ResNet-101	52.6
ANN [69]	ResNet101	52.8
CPNet [58]	ResNet-101	53.9
CFNet [64]	ResNet-101	54.0
ACNet [24]	ResNet-101	54.1
APCNet [26]	ResNet101	54.7
DNL [57]	HRNetV2-W48	55.3
DRANet [22]	ResNet-101	55.4
OCR [60]	HRNetV2-W48	56.2
SETR-L MLA [67]	ViT-L/16	55.8
Seg-B [†] /16	DeiT-B/16	53.9
Seg-B [†] -Mask/16	DeiT-B/16	55.0
Seg-L/16	ViT-L/16	56.5
Seg-L-Mask/16	ViT-L/16	59.0

Ablation study: SOTA comparison on Cityscapes

Method	Backbone	mIoU (MS)
PSANet [66]	ResNet-101	79.1
DeepLabv3+ [10]	Xception-71	79.6
ANN [69]	ResNet-101	79.9
MDEQ [5]	MDEQ	80.3
DeepLabv3+ [10]	ResNeSt-101	80.4
DNL [57]	ResNet-101	80.5
CCNet [31]	ResNet-101	81.3
Panoptic-Deeplab [12]	Xception-71	81.5
DeepLabv3+ [10]	ResNeSt-200	82.7
SETR-L PUP [67]	ViT-L/16	82.2
Seg-B ⁺ /16	DeiT-B/16	80.5
Seg-B ⁺ -Mask/16	DeiT-B/16	80.6
Seg-L/16	ViT-L/16	80.7
Seg-L-Mask/16	ViT-L/16	81.3



Ablation study: performance



Conclusions

- introduces a **pure transformer approach** for semantic segmentation
- relies on the **encoding of all images patches** instead of a single cls token
- the sequence-to-sequence structure is **highly effective at capturing global context** across the entire image
- the simple point-wise linear decoder applied directly to the patch encodings is shown to achieve excellent baseline performance
- decoding with a mask transformer further improves the performance
- first step toward a **unified model** for semantic, instance and panoptic segmentation

Experiment description

- 2 different fine-tuning of segmenter tiny on cityscapes:
 - one with **mask transformer decoder**
 - the other with **linear decoder**
- same training configurations
- comparison between the 2 resulting models
- comparison with the paper's results on cityscapes

Training configurations

- image size: 1024
- crop size: 768
- window size: 768
- window stride: 512
- epochs: 60
- batch size: 4
- lr scheduler: polynomial
- optimizer: SGD

Model configurations

- patch size: 16
- encoder layers: 12
- mask decoder layers: 2
- heads: 3
- embedding dim: 192

Metrics

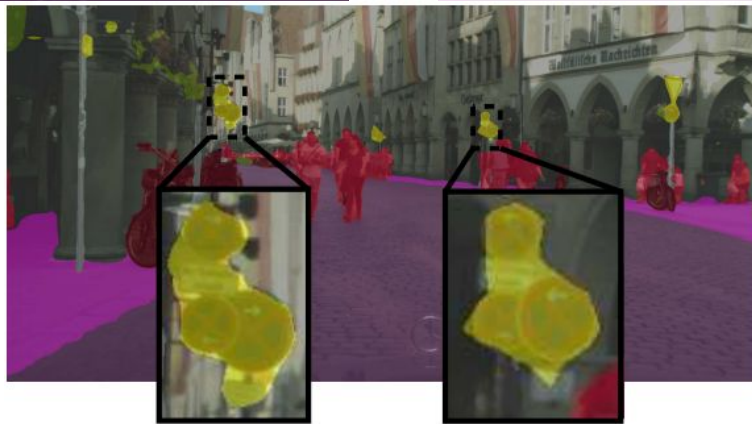
Model	mIoU (SS/MS)	aAcc (SS/MS)	mAcc (SS/MS)
Seg-L/16	- / 80.7	-	-
Seg-L-Mask/16	79.1 / 81.3	-	-
Seg-Ti/16	52.8 / 54.3	91.8 / 92.4	61.9 / 62.2
Seg-Ti-Mask/16	61.8 / 64.9	93.8 / 94.4	72.0 / 73.7

Visual results

Seg-Ti-Mask/16



Seg-Ti/16



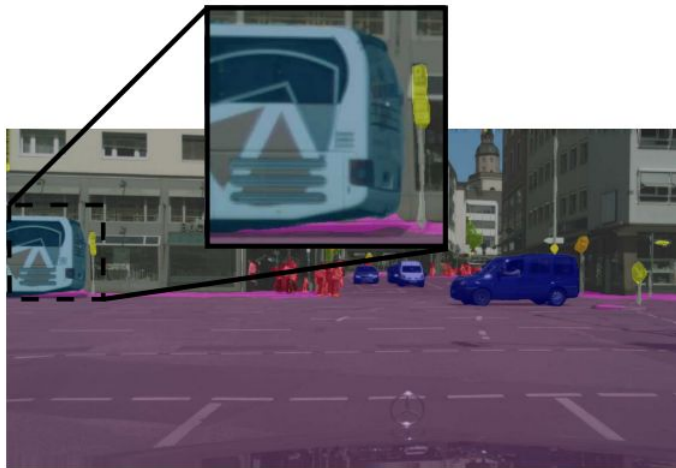
Seg-L-Mask/16

Visual results

Seg-Ti-Mask/16



Seg-Ti/16



Seg-L-Mask/16